

The entropy cost of a linear log-concavity failure run

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Abstract

The degree-free forest obstruction extends to graphs whose cyclic part has subpolynomially many independent configurations. Define $\kappa(G) = \min_{G-S \text{ a forest}} \log_2 Z_{G[S]}(1)$. If $\kappa(G_n) = o(\log n)$, every consecutive strict log-concavity failure run in Z_{G_n} has length $o(n)$. In particular, this holds when the feedback vertex number is $o(\log n)$, and when deleting a clique of size $n^{o(1)}$ leaves a forest. Edges incident with the deleted part are unrestricted. The proof uses an exact marginal reweighting identity, not an assumption about the conditional independence of cyclic vertices.

Scope. This is a proved extension of the supplied forest argument, rederived below with an optimized variance exponent. It is a research draft: independent review and publication priority are not established. The accompanying exact finite audit checks the new transfer identities; enumeration is not a substitute for the universal proof.

1 Statement

All graphs are finite and simple. Write

$$Z_G(\lambda) = \sum_{I \in \mathcal{I}(G)} \lambda^{|I|} = \sum_{j=0}^{\alpha(G)} i_j(G) \lambda^j.$$

A strict failure at index j means $i_j^2 < i_{j-1} i_{j+1}$. A run at $u, \dots, v$ has length $L = v - u + 1$; put $d = L + 1$. Set

$$\kappa(G) = \min_{S \subseteq V(G): G-S \text{ is a forest}} \log_2 Z_{G[S]}(1).$$

The empty graph counts as a forest, and $Z_\emptyset = 1$.

Theorem 1 (An explicit structural obstruction). *Fix $0 < \delta \leq 1$, and define*

$$\Lambda_\delta = 4^{1/\delta}, \quad B_\delta = 2\sqrt{\Lambda_\delta}, \quad \eta_\delta = \frac{\log(1 + 1/\Lambda_\delta)}{\log B_\delta + \log(1 + 1/\Lambda_\delta)} > 0.$$

If an n -vertex graph has a strict failure run with $L + 1 \geq \delta n$, then

$$\boxed{\kappa(G) > \min \left\{ \frac{\delta n}{\sqrt{12}}, \frac{\eta_\delta \log n - \log(36/\delta^2)}{\log(2\Lambda_\delta)} \right\}.} \quad (1)$$

Here and below unmarked logarithms are natural. In particular, for every fixed δ , a run of this length forces $\kappa(G) \geq c_\delta \log n$ for all sufficiently large n , where $c_\delta = \eta_\delta / [2 \log(2\Lambda_\delta)]$.

Corollary 2. *For any sequence of graphs of orders $n \rightarrow \infty$,*

$$\kappa(G_n) = o(\log n) \implies \frac{\text{longest strict failure run in } G_n}{n} \longrightarrow 0.$$

There are no maximum-degree, edge-count, or component-count assumptions.

2 The exact transfer identity

Allow positive vertex activities λ_v . For any graph H , $Z_H = \sum_{I \in \mathcal{I}(H)} \prod_{v \in I} \lambda_v$. Fix any $S \subseteq V(G)$, let $F = G - S$, and let P_F denote the hard-core measure on F . For $J \in \mathcal{I}(F)$, define

$$W(J) = Z_{G[S \setminus N_G(J)]}, \quad C = Z_{G[S]}.$$

Every independent set in G is uniquely $J \cup K$, where $K \in \mathcal{I}(G[S \setminus N_G(J)])$. Consequently

$$P_G(I \cap V(F) = J) = \frac{P_F(J)W(J)}{\mathbb{E}_F W}, \quad 1 \leq W(J) \leq C. \quad (2)$$

This is an exact identity. No acyclicity is needed here.

Lemma 3 (Variance transfer). *Let $X_G = |I|$, $X_F = |J|$, and $a = \alpha(G[S])$. Then*

$$\sqrt{\text{Var}_G X_G} \leq \sqrt{C \text{Var}_F X_F} + \frac{a}{2}. \quad (3)$$

More generally, every nonnegative observable $f(J)$ satisfies $\mathbb{E}_G f(I \cap F) \leq C \mathbb{E}_F f(J)$.

Proof. The expectation inequality follows from (2), since $\mathbb{E}_F W \geq 1$. Under P_G , write $X_G = Y + T$, with $Y = |I \cap F|$ and $0 \leq T = |I \cap S| \leq a$. Choosing $b = \mathbb{E}_F X_F$ gives

$$\text{Var}_G Y = \min_{t \in \mathbb{R}} \mathbb{E}_G (Y - t)^2 \leq \mathbb{E}_G (Y - b)^2 \leq C \text{Var}_F X_F.$$

Also $\text{Var}_G T \leq a^2/4$: a variable in $[0, a]$ has variance at most $a^2/4$. The triangle inequality in $L^2(P_G)$ for the centered variables Y and T proves (3). These variables need not be independent. $\square$

3 A self-contained forest variance estimate

Lemma 4. *Let F be a forest with at most $n \geq 1$ vertices and positive activities at most $\Lambda \geq 1$. Put*

$$B = 2\sqrt{\Lambda}, \quad q = \frac{\Lambda}{1 + \Lambda}, \quad \eta = \frac{\log(1/q)}{\log(B/q)}.$$

Then, without a lower threshold on n ,

$$\text{Var}_F X_F \leq \frac{3}{4} n^{2-\eta}. \quad (4)$$

Proof. The empty forest is immediate. Otherwise let S_v indicate occupancy, $\mu_v = \mathbb{E} S_v$, $s_v = \sqrt{\text{Var } S_v}$, and $w_v = \mu_v/(1 - \mu_v)$. For an oriented edge $u \rightarrow v$, let $R_{u \rightarrow v}$ be the occupancy odds at u in the component obtained by deleting uv . Conditioning at its root gives

$$R_{u \rightarrow v} = \lambda_u \prod_{z \sim u, z \neq v} (1 + R_{z \rightarrow u})^{-1}, \quad w_v = \lambda_v \prod_{u \sim v} (1 + R_{u \rightarrow v})^{-1}.$$

In particular $R_{u \rightarrow v} \leq \Lambda$. Adjacent vertices cannot both be occupied, so, with $a_{uv} = |\text{Corr}(S_u, S_v)|$,

$$a_{uv}^2 = w_u w_v.$$

Fix v and write $x_u = R_{u \rightarrow v}$. Edge conditioning gives $w_u = x_u/(1 + R_{v \rightarrow u}) \leq x_u$. Therefore

$$\sum_{u \sim v} a_{uv}^2 \leq \lambda_v \frac{\sum_{u \sim v} x_u}{\prod_{u \sim v} (1 + x_u)} \leq \lambda_v \leq \Lambda.$$

The middle product is at least $1 + \sum x_u$. For an edge, writing $R = R_{u \rightarrow v}$ and $T = R_{v \rightarrow u}$ also gives

$$a_{uv}^2 = \frac{R}{1 + R} \frac{T}{1 + T} \leq q^2.$$

Let A be weighted adjacency with entries a_{uv} . Root each component, and let D contain the parent-to-child entries of A . Distinct rows of D have disjoint supports. Thus $DD^\top$ is diagonal, with entries at most Λ , and $\|A\|_2 = \|D + D^\top\|_2 \leq 2\sqrt{\Lambda} = B$.

For vertices at distance r in the same component, the tree Markov property implies

$$\text{Cov}(S_u, S_v) = (-1)^r s_u s_v \prod_{e \in P(u,v)} a_e.$$

For completeness, a centered function of a binary variable is a scalar multiple of its centered indicator. Hence on an edge the conditional expectation of the normalized centered indicator is its correlation times the normalized centered indicator at the other endpoint. Composing this relation along the path proves the formula. Distinct components are independent.

The simple path is one of the nonnegative walks counted in A^r . Writing $y_v = s_v$, the sum over ordered pairs at distance r is at most $y^\top A^r y \leq (n/4)B^r$. Each individual pair at distance r has absolute covariance at most $q^r/4$. Therefore, for every integer $R \geq 0$,

$$\text{Var}_F X_F \leq \frac{n}{4} \sum_{r=0}^R B^r + \frac{n^2}{4} q^{R+1} \leq \frac{n}{2} B^R + \frac{n^2}{4} q^{R+1}. \quad (5)$$

The exponent $R+1$ uses the fact that all remaining distances exceed R . Put $t = \log n / \log(B/q)$ and $R = \lfloor t \rfloor$. Then

$$B^R \leq B^t = n^{1-\eta}, \quad q^{R+1} \leq q^t = n^{-\eta}.$$

Substitution in (5) proves (4). $\square$

4 A failure window and its centered activity

Lemma 5. *For any n -vertex graph with a strict failure run at $u, \dots, v$, put $c = (u+v)/2$ and $d = v-u+2$. There is a unique common activity $\lambda > 0$ with $\mathbb{E}_\lambda X_G = c$, and at that activity*

$$\text{Var}_\lambda X_G > \frac{d^2}{12}, \quad \lambda \leq 2^{2n/d}. \quad (6)$$

Proof. Every coefficient from 0 through $\alpha(G)$ is positive, since subsets of a maximum independent set are independent. The mean tends to 0 and $\alpha(G)$ as λ tends to 0 and ∞ , respectively. Its derivative with respect to $\log \lambda$ is the positive variance. Also $0 < c < \alpha(G)$.

At the centered activity let $p_j = i_j \lambda^j / Z_G(\lambda)$. Tilting preserves the strict failure inequalities. The arithmetic-geometric mean inequality implies $p_{j-1} - 2p_j + p_{j+1} > 0$ for $u \leq j \leq v$. For any numbers $z_0, \dots, z_d$, collecting coefficients gives the exact identity

$$\sum_{j=0}^d \left[(j-d/2)^2 - \frac{d(d+2)}{12} \right] z_j = \sum_{j=1}^{d-1} \frac{j(j+1)(d-j)(d-j+1)}{12} (z_{j-1} - 2z_j + z_{j+1}).$$

Take $z_j = p_{u-1+j}$. The right side is nonnegative, and every index outside the window has squared distance from c larger than $d(d+2)/12$. Summing all probabilities gives $\text{Var} X_G \geq d(d+2)/12 > d^2/12$.

The Shannon entropy of the independent-set distribution, in nats, is $H = \log Z_G(\lambda) - c \log \lambda \leq n \log 2$. Since $Z_G(\lambda) \geq \lambda^{\alpha(G)}$, $(\alpha(G) - c) \log \lambda \leq n \log 2$. The enlarged window ends at $v+1 \leq \alpha(G)$, so $\alpha(G) - c \geq d/2$. If $\log \lambda \geq 0$ this yields the stated bound; if $\lambda < 1$, the bound is automatic. $\square$

5 Proof of the entropy barrier

Proof of Theorem 1. Choose any S with $F = G - S$ a forest, and put $h = \log_2 Z_{G[S]}(1)$ and $a = \alpha(G[S])$. All subsets of a maximum independent set in $G[S]$ contribute to $Z_{G[S]}(1)$, so $a \leq h$. At the centered activity of Lemma 5, $\lambda \leq \Lambda_\delta$. Consequently

$$C = Z_{G[S]}(\lambda) \leq Z_{G[S]}(1) \Lambda_\delta^a \leq (2\Lambda_\delta)^h. \quad (7)$$

Apply Lemmas 3 and 4, using the ambient order n as an upper bound on the forest order. We obtain

$$\frac{\delta}{\sqrt{12}} < \frac{\sqrt{\text{Var}_G X_G}}{n} \leq \frac{\sqrt{3}}{2} (2\Lambda_\delta)^{h/2} n^{-\eta_\delta/2} + \frac{h}{2n}. \quad (8)$$

If $h > \delta n / \sqrt{12}$, the desired minimum bound follows. Otherwise the last term is at most $\delta / (2\sqrt{12})$. Rearranging and squaring the remaining positive terms gives

$$(2\Lambda_\delta)^h > \frac{\delta^2}{36} n^{\eta_\delta}.$$

Taking logarithms proves the other alternative in (1). The bound holds for every admissible S , including one attaining the minimum defining $\kappa(G)$. For fixed δ , the logarithmic term is eventually the smaller term and eventually at least $c_\delta \log n$. $\square$

Proof of Corollary 2. If the conclusion failed, some fixed $\delta > 0$ and an infinite subsequence would have $L + 1 \geq \delta n$. Theorem 1 would force $\kappa(G_n) \geq c_\delta \log n$ along that subsequence, contradicting $\kappa(G_n) = o(\log n)$. $\square$

6 Concrete consequences

Corollary 6 (Feedback vertices and a finite threshold). *Suppose deleting k vertices makes G a forest. Fix δ and use the parameters of Theorem 1. A run with $L + 1 \geq \delta n$ is impossible whenever*

$$n \geq \max \left\{ 1, \frac{\sqrt{12}k}{\delta}, \left\lceil \frac{36(1 + \Lambda_\delta)^k}{\delta^2} \right\rceil^{1/\eta_\delta} \right\}.$$

Thus feedback vertex number $o(\log n)$ suffices for the asymptotic obstruction.

Proof. For the chosen S , $a \leq k$ and $C \leq (1 + \Lambda_\delta)^k$. Under the displayed threshold each of the two upper terms in the analogue of (8) is at most $\delta / (2\sqrt{12})$, contradicting its strict lower bound. Also $Z_{G[S]}(1) \leq 2^k$, so $\kappa(G) \leq k$. $\square$

Large dense cores. If S induces a clique of size s , then $Z_{G[S]}(1) = 1 + s$. Thus a clique of size $s = n^{o(1)}$ may be deleted, with every edge between the clique and the remaining forest arbitrary, while the conclusion still holds. For example, $s = \lceil \exp(\sqrt{\log n}) \rceil$ is far larger than $\log n$, but its configuration cost is only $O(\sqrt{\log n})$. More generally, $Z_{G[S]}(1) = n^{o(1)}$ is the sufficient condition.

What has changed from the supplied proof. The supplied note treats forests. The new structural statement treats arbitrary attachments to an induced cyclic core, charging that core by the number of its independent configurations. The forest exponent is also optimized: balancing B^R against q^{R+1} yields $\eta = \log(1/q) / \log(B/q)$ for every ambient order $n \geq 1$. Neither the exact marginal identity nor the graph entropy barrier relies on the finite audit.

Verification. Run `python verify_core_entropy_barrier.py`. It uses the standard library and exact rational arithmetic. It independently enumerates graph independent sets, verifies the reweighting identity, observable domination, the sharp square-root variance inequality without floating-point roots, and the core counting inequalities. The JSON report states the finite scope and the number of checks.

Literature boundary. Spectral estimates on trees below e^2 appear in X. Chen, X. Yang, Y. Yin and X. Zhang, *Spectral Independence Beyond Total Influence on Trees and Related Graphs*, [arXiv:2404.04668](https://arxiv.org/abs/2404.04668). The supplied *A degree-free obstruction to linear runs of log-concavity failures in forests* (28 September 2026) is the starting point for this note. No claim that the present extension is absent from all existing literature is made.